MILITARY SPECIFICATION

RESISTORS, FIXED, FILM,
ESTABLISHED RELIABILITY
STYLE RNR50[1/]

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RNR50[1/], established reliability, film, fixed resistors.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification. The following specification forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of this document shall be that listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-55182 - Resistors, Fixed, Film, Established Reliability, General Specification for.

(Copies of the specification, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-55182, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

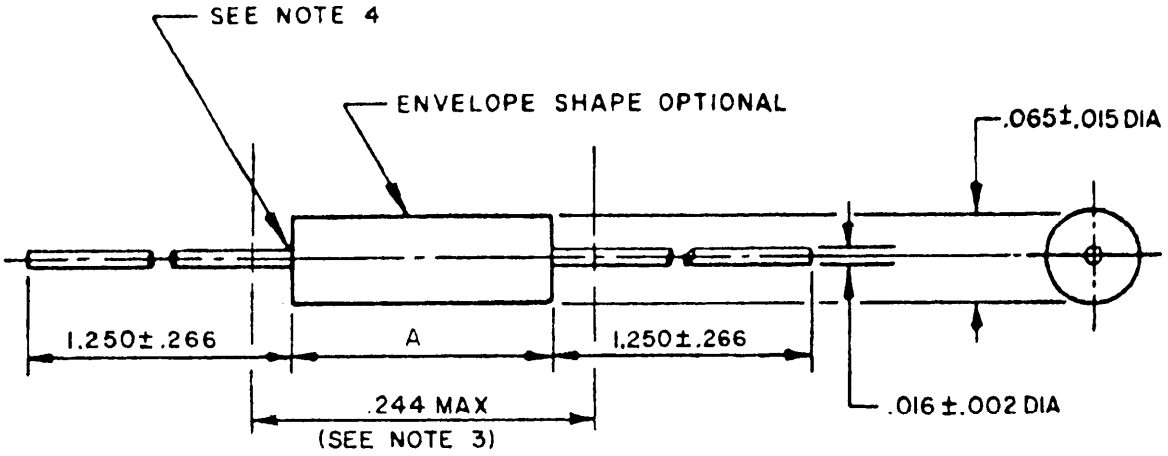
3.2.1 Characteristic. Style RNR50 is available in characteristics C, E, H, J, and K.

3.2.2 Terminal. Terminal types RNR, RNC, and RNN are available in all characteristics. Terminal RNR is inactive for design with characteristics H, J, and K.

1/ Third letter is variable, dependent upon lead material or capability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, NJ 07703 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Characteristic	A ±.020
C, E	.180
H, J, K	.150

INCHES	MM
.002	0.05
.015	0.38
.016	0.41
.020	0.51
.065	1.65
.150	3.81
.180	4.57
.244	6.20
.266	6.76
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Maximum length is clean lead to clean lead.
4. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.

FIGURE 1. Style RNR50 resistors. 1/

1/ See footnote 1, page 1.

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3.3 Power rating. The power rating shall be 0.05 watt (see 6.2).

3.4 Voltage rating. The continuous working voltage shall not exceed 200 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows:

Minimum resistance:

Characteristic:

C, E, H, J, K	10.0 ohms.

Maximum resistance:

Characteristic

C, E	0.1 megohm.
H, J, K	0.796 megohm.

3.6 Voltage coefficient (applicable to resistors of 1,000 ohms and above). The voltage coefficient shall not exceed ±0.005 percent per volt.

* 3.7 Insulation. Insulation material surrounding the resistor body shall extend around the entire shoulder of each end cap toward the axis of the lead attachments. Complete back wall coverage is optional.

* 3.8 Mechanical shear. The mechanical shear test is applicable to this specification.

* 3.9 Marking. Due to size limitations this style resistor shall be marked with the following minimum information:

042K - Date code, characteristic.
10R0 - Coded resistance value.
FMCJ - Tolerance, failure rate, lead material, JAN marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. Sampling and inspection shall be in accordance with MIL-R-55182.

* 4.2 Qualification conformance inspection. In addition to the quality conformance inspection required by MIL-R-55182, verify compliance with 3.7 on a sample size equivalent to the sample required for a visual inspection of group A, subgroup 2. AQL and LQ shall be defined for a major defect for this requirement.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-55182.

5.2 Marking. Package shall be marked with assigned part number, manufacturers identification name, logo or code number (FSCM).

5. NOTES

6.1 Notes. The notes specified specified in MIL-R-55182 are applicable to this specification.

6.2 Power rating at 70°C. The power rating at 70°C for 2,000 hours life test has been established as 0.1 watt, maximum voltage 200 volts. It should be noted that the failure rate level is established at the single condition of 125°C, 0.05 watt and 10,000 hour life test duration.

6.3 Supersession data. This specification supersedes MIL-R-38101/20(USAF).

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Lead: Tolerance: See table I.

6.4 MIL-R-10509 substitution data. Resistors of this specification, regardless of their failure rate designation, are substitutes for resistors of the same resistance value and tolerance specified in the inactivated specification MIL-R-10509/8. Substitute characteristics and terminal types are indicated in table I.

6.5 MIL-R-55182 substitution data. Resistors of this specification, regardless of their failure rate designation, are substitutes for resistors of the same resistance value and tolerance specified in the inactivated specification MIL-R-55182/11.

TABLE I. Characteristic and terminal substitution.

MIL-R-10509 Style RN50	MIL-R-55182/7 Substitute characteristic	MIL-R-55182/7 Substitute terminal
Inactive characteristic		
C	H	RNC
E	J	RNC

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - ER, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 80, 85
DLA - ES

User activities:
Army - AT, ME, AV
Navy - MC, CG
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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